

Tyler Wallace

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Summary

Software engineer with 9+ years of experience building real-time 3D applications, VR training systems, and full-stack platforms for defense and commercial clients. Shipped Unity SDKs, multiplayer XR applications, and mission-critical intelligence platforms across Air Force and private sector environments. Actively transitioning into AI engineering — pursuing a Bachelor's in Artificial Intelligence with a focus on machine learning and intelligent systems development. Previous Top Secret clearance holder eligible for renewal, offering a significantly faster reinstatement process than clearing a new candidate.

Skills

Languages:

C# • JavaScript • PHP • HTML5 • CSS • SQL

Platforms & Engines:

Unity • Meta Quest 3 • Vive Focus 3 • WebGL

Tools & Libraries:

Git • NPM • REST APIs • MySQL • Bootstrap • jQuery

Methodologies:

Agile • SDK Development • Software Lifecycle Management • Requirements Gathering

Currently Learning:

Python • Machine Learning • Unity ML-Agents • OpenAI API • Computer Vision

Experience

Software Engineer 04/2022 – 05/2025

Dynepic, Inc – Remote

- Developed Unity SDKs targeting Meta Quest 3 and Vive Focus 3 to streamline training application creation and digital asset integration across client projects.
- Engineered VR device authentication systems providing secure, direct access to training materials in compliance with defense client requirements.
- Built multiplayer mixed reality applications showcasing advanced platform features including real-time asset streaming across connected devices.
- Contributed to the MOTAR platform — an AI-assisted training system that reduced a 27-day Air Force training course to an average of 12.5 days (46% reduction) while maintaining equivalent learning outcomes.
- Optimized VR gameplay experiences across diverse user skill levels, tailoring interactions for both novice and experienced personnel.
- Operated as a solo developer within a small XR team, owning full feature development cycles from architecture through deployment via NPM.
- Presented live technical demos to stakeholders and delivered user training sessions on application capabilities.
- Implemented software lifecycle best practices including Git collaboration, automated testing, and versioned code releases.

Software Developer 03/2020 – 04/2022

United States Air Force – Hill AFB, UT

- Served as lead developer on a small team to build proof-of-concept VR training simulations demonstrating that internal Air Force developers could independently create high-quality training applications.
- Engineered Unity and C# simulations of aircraft maintenance tasks, providing immersive procedural training for Air Force personnel.
- Trained 2 incoming Airmen in object-oriented design, 3D application development, and Air Force software development standards.

- Designed modular, reusable code architecture enabling simulations to integrate across multiple training scenarios and platforms.

Operations Programmer 02/2017 – 03/2020

United States Air Force – Fort Meade, MD

- Sole developer for 2 years on a mission-critical full-stack intelligence platform serving an active Intelligence Support Squadron, later growing the team by onboarding an additional developer.
- Replaced a legacy Excel-based workflow with a purpose-built web platform, dramatically reducing data entry and retrieval time for approximately 100 active users operating 24/7.
- Conducted interviews with linguists and training managers to derive requirements, identify pain points, and continuously improve platform capabilities.
- Engineered analytical dashboards enabling commanders to identify intelligence collection trends and make data-driven operational decisions.
- Built full-stack using PHP, HTML5, CSS, JavaScript, jQuery, MySQL, and Bootstrap with an emphasis on usability and mission-critical reliability.
- Collaborated directly with Subject Matter Experts (SMEs) to ensure platform accuracy, relevance, and alignment with operational needs.

Notable Projects

MOTAR — Dynepic, Inc | *Unity • C# • VR • Meta Quest 3 • Vive Focus 3*

- Contributed to Dynepic's MOTAR platform, a Learning Content Management System integrating VR, AR, AI, and immersive web technologies to deliver scalable Air Force technical training.
- Platform reduced a 27-day Crew Chief Fundamentals course to 12.5 days on average — a 46% reduction — while achieving knowledge test scores within 7–8% of traditional instruction and 93.5% of trainees meeting or exceeding performance standards.
- Delivered 170+ lessons across 4 proficiency levels using 9 distinct media types, validating immersive technology as a viable replacement for traditional classroom instruction.

Intelligence Analytics Platform — USAF | *PHP • JavaScript • MySQL • HTML5 • CSS*

- Designed and built a full-stack web platform from the ground up as the sole developer, replacing a cumbersome Excel-based workflow for an active Intelligence Support Squadron.
- System supported ~100 users operating 24/7, enabling linguists to efficiently enter, retrieve, and analyze intelligence data with real-time trend analytics for operational decision-making.

VR Maintenance Training Suite — USAF | *Unity • C# • VR*

- Led development of proof-of-concept VR simulations replicating aircraft maintenance procedures, demonstrating the viability of in-house Air Force XR training development.
- Established development standards and mentored team members, creating a reusable framework adopted across multiple training scenarios.

Education

Bachelor's Degree, Artificial Intelligence — In Progress

Grand Canyon University – Online | Expected 2027

Associate's Degree, Computer Science

Community College of the Air Force – Montgomery, AL | January 2019

Security Clearance & Certifications

Previous Top Secret Clearance Holder – Eligible for renewal | Faster reinstatement than new candidate.

CompTIA Security+ – Previously certified | Prepared to recertify.